

JANUS PHOTONIC COMPUTING ARCHITECTURE

Consolidated Design Record

Physics Verification, Amplifier Design, and Fabrication Supply Chain

Patent Application No. 202611052791 (Patent Pending)

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Status: Amplifier stage locked to Si_3N_4 (foundry-constrained) · Core timing/power architecture verified

Source documents merged into this record:

- JANUS Architecture — Physics-Level Error Analysis, Corrections & Verified Solutions (June 2026)
- GaP Parametric Amplifier: Material Selection, Design Derivation & Power Budget (July 2026)
- Photonic Amplification & Fabrication Design Log — Si_3N_4 Verification and Supply Chain (July 2026)

Executive Summary

This record consolidates three working documents into a single, continuous account of the JANUS photonic-electronic accelerator's verified physics, its amplifier stage design history, and its fabrication supply chain. It is organized final-state-first: Part I states the current locked configuration exactly as it should be read today. Part II preserves the full derivation trail — the physics error analysis that produced the current timing/power model, the complete GaP amplifier investigation, and the Si_3N_4 paths that were tested and rejected — so that no reasoning, rejected alternative, or intermediate number from the original three documents is lost. Part III merges all outstanding open items. Part IV merges all references.

Current Status at a Glance

Subsystem	Current Locked State	Basis
Core timing model	~100 GHz optical clock, EAM-limited, wave-pipelined	Error Analysis §2
GaP microring spectral match (legacy)	Superseded — no longer the amplifier path (see below)	Error Analysis §3; GaP doc
Detector signal / margin (33GHz analysis)	+2.5 dBm signal, +34.26 dB margin (narrative-only fix)	Error Analysis §4
Amplifier stage — FINAL	Si_3N_4 ring cascade, $R=30\mu\text{m}$, $Q=3\times 10^6$, 11 rings, 30.94–31.66 dB gain, CMT-simulation verified	Fabrication Design Log §1
Amplifier stage — leading future candidate	Pure GaP microring, $R=5\mu\text{m}$, $Q\approx 290,000$ –300,000, 21 dB — not currently manufacturable at required Q via MPW	GaP doc §11
Noise floor (100 GHz, real detector data)	–17.2 dBm (rigorous recalculation, supersedes earlier –31.76 dBm and –27 dBm estimates)	Fabrication Design Log §3
Laser source	CW InP-on-Si + TFLN MZM + chirped Bragg grating, ~35–40% WPE	Error Analysis §6
Laser power budget (datacenter)	~0.44–0.66 W (down from erroneous 13.33 W)	Error Analysis §5
CMOS backend	21 parallel CRT pipelines at ~4.76 GHz (TSMC N3)	Error Analysis §7
Si_3N_4 photonic backbone foundry	LIGENTEC (Switzerland), AN800 process, via VLC Photonics for PDK support	Fabrication Design Log §4.1
CMOS foundry	SCL Mohali (180nm live) / Dholera (targeting 28–90nm) via C2S	Fabrication Design Log §4.2
GST/PCM switches	LioniX International (Netherlands) via Europractice — heater only, GST deposition is a separate rented-cleanroom step	Fabrication Design Log §4.3
Die-to-die bonding	imec (Belgium) — true D2D pitch/yield still unverified	Fabrication Design Log §4.4

Why Si_3N_4 Over GaP, For Now

GaP remains the physically stronger candidate for the amplifier stage — its Kerr index and demonstrated integration make it the design that, on paper, offers the more favorable pump-power/gain trade at small radius. However, the GaP microring specification ($R=5\mu\text{m}$, $Q\approx 290,000$ –300,000) requires a loaded Q that is not currently obtainable through any MPW (multi-project wafer) run accessible to this project. No foundry in the current supply chain (Section 4 of Part I)

offers GaP-on-insulator rings at that Q. Si_3N_4 , by contrast, is directly available at LIGENTEC with $Q > 2 \times 10^7$ on a qualified, automotive-grade production line — comfortably above the $Q = 3 \times 10^6$ the locked design actually uses. The decision rule going forward: Si_3N_4 is the design-of-record while GaP at the required Q remains foundry-unavailable; if GaP-on-insulator at $Q \geq 290,000$ becomes accessible via MPW in the future, GaP is the preferred upgrade path over Si_3N_4 .

PART I — Current Locked Specification

This part states the JANUS design as it stands today, incorporating every correction and decision from all three source documents. Historical derivations, rejected alternatives, and the full reasoning trail are preserved in Part II.

1. Corrected Core Architecture (Timing, Detection, Laser, CMOS)

These parameters were established in the June 2026 physics-level error analysis and remain current; they are unaffected by the later amplifier-stage material decision (Si_3N_4 vs. GaP).

Parameter	Original Paper	Corrected Value	Source
Operating frequency	33 GHz	~100 GHz (optical), EAM-limited	Error 1
Timing model	Serial sum: $t_{\text{EAM}} + t_{\text{PD}} + t_{\text{wire}} + t_{\text{guard}}$	Wave-pipelined, throughput set by EAM only	Error 1
Guard band term (t_{guard})	18.6 ps per cycle	0 (amortised once per ~0.1ms epoch)	Error 1
Detector signal power	-7.5 dBm (narrative)	+2.5 dBm (correct, matches Table VIII)	Error 3
Detection margin	+24.26 dB	+34.26 dB	Error 3
Per-laser time-avg power	1.6 mW	0.032 mW	Error 4
Datacenter laser electrical power	13.33 W	~0.44–0.66 W	Error 4
Laser source type	Mode-locked laser (MLL) at 33 GHz	CW InP-on-Si + TFLN MZM + chirped Bragg grating	Error 5
Laser WPE	60% (physically unreachable for MLL)	35–40% (demonstrated, CW)	Error 5
Pulse width	1.5 ps	5 ps	Error 2 (superseded downstream — see §1.3 below)
CMOS backend pipelines	1 (bottleneck undefined)	21 parallel at ~4.76 GHz, TSMC N3	Error 6
DC throughput (64-bit exact)	919.2 TMAC/s	~2,785 TMAC/s	Error 1
CMOS dynamic power	Baseline	~2.4× lower (voltage scaling from parallelism)	Error 6

Unaffected core innovations: One-Hot spatial encoding, non-volatile PCM Beneš routing, single-wavelength operation (FWM/XPM suppression), CRT-based fault visibility, Z-axis thermal partitioning, and spatial RNS decomposition are all confirmed physically valid and untouched by any of the corrections below.

1.1 Wave-Pipelined Timing Model — Summary

The architecture's throughput ceiling is set purely by EAM modulation bandwidth (10 ps → 100 GHz), not by summing every latency in the chain. Detector recovery (0.60 ps), wire propagation (1.33 ps, latency-only), Beneš routing (750 ps, pipelined), and PCM settling (~0.1 ms, amortised per epoch, not per-cycle) do not limit throughput. End-to-end single-computation latency is 750 ps (Beneš) + 210 ps (CRT reconstruction) = 960 ps, i.e. ~10 pipeline stages at the corrected 100 GHz clock.

1.2 Detection Margin

Tracing the optical chain (laser → 1:64 split – 18.06 dB → GaP/parametric front-end +30 dB → Beneš routing –7.50 dB) gives a net +4.44 dB offset from the laser, and a detector signal of +2.5 dBm rather than the –7.5 dBm stated in the original narrative (Table VIII was already internally correct). Against the corrected noise floor of –17.2 dBm (Part I §2.3, superseding the original –31.76 dBm 33 GHz figure), this is a narrative-level correction only — the underlying power budget in the paper's own tables did not need to change.

1.3 Laser Source

Source: CW InP-on-Si laser at 1550 nm (30–40% WPE, use 40% as design point). Modulator: TFLN Mach-Zehnder Modulator at 100 GHz, $V_\pi < 1V$, insertion loss 2.5 dB. Pulse shaping: chirped Bragg grating compressor, insertion loss 1.5 dB, producing ~5 ps output pulses. Combined effective system WPE after insertion losses: ~35%. This replaces the mode-locked laser (MLL) assumption entirely — 60% WPE is demonstrated only for CW sources; real GHz-repetition-rate MLLs achieve 2–10% WPE, which would have corrupted the full power budget by 6–30× if left uncorrected.

1.4 CMOS Backend

At the corrected 100 GHz optical rate, results arrive every 10 ps but the CRT reconstruction pipeline has ~210 ps latency — no single CMOS pipeline at TSMC N3 can close timing at 100 GHz for multi-bit adder trees. The fix, standard in high-speed ADC and 100G/400G SerDes design, is time-interleaving: 21 parallel CRT reconstruction pipelines, each running at a comfortable ~4.76 GHz, fed by a 1:21 electronic DMUX (SiGe BiCMOS or InP HBT) after the TIA. Each pipeline is tagged by computation ID; since all residue tiles for one computation share the optical master clock, they arrive in the same 10 ps window and route together with no out-of-order reconstruction problem. This also yields a ~2.4× dynamic-power advantage over a hypothetical (impossible) single 100 GHz pipeline, because 21 pipelines at $V_{DD} \approx 0.65V$ beat one pipeline that would require $V_{DD} > 1.0V$.

2. Final Amplifier Design — Si₃N₄ Ring Cascade

This is the current design-of-record for the JANUS optical amplification stage, replacing all earlier GaP-based specifications (superseded designs preserved in Part II).

Verified Total Gain: 30.94 – 31.66 dB	
Simulation-confirmed. Margin above real detector noise floor: +9.14 to +9.86 dB.	
Parameter	Value
Ring radius (R)	30 μm
Loaded Q factor	3×10^6
Cascaded rings	11
Pump power per ring	1.0 – 1.5 mW
Total amplifier pump budget	11 – 16.5 mW
Real ring input (post-split)	–39 dBm
Final output signal	–8.06 dBm / –7.34 dBm
Detector noise floor (100 GHz)	–17.2 dBm
Stability, all 11 stages	CV ≈ 0.00% · drift ≈ 0.00%

2.1 Verification Method

Three-mode coupled-mode theory (CMT) simulation — pump, signal, idler — numerically integrated (SciPy solve_ivp, RK45, rtol=1e-9). The threshold formula used was independently cross-validated against a real published measurement before being trusted for this design: a SiC microring OPO where the formula predicted 2.71 mW against a measured 2.7 mW.

Stability criterion required both: (1) signal photon number settling to under 5% coefficient of variation, AND (2) under 2% drift between two independent late-time windows. The second, stricter check was added specifically after a looser single-window check produced a false “stable” verdict that a longer integration run exposed as a bug.

Scope of this result: this is a first-principles simulation, not measured hardware. It does not include thermal effects, free-carrier dynamics, or fabrication mismatch beyond a basic phase-error test. Treat as strong directional evidence — real device characterization is still required before committing to fabrication.

2.2 Noise Floor — Final Verified Number

Figure	Value	Status
Real noise floor used (100 GHz)	–17.2 dBm	Verified
JANUS paper's own claimed figure (33 GHz)	–31.76 dBm	Original claim
Naive bandwidth-scaling to 100 GHz	~~27 dBm	First-pass estimate
Rigorous recalculation, real detector data	–17.2 dBm	Used

Derivation: thermal-noise-limited receiver, 50Ω load, T=300K, using the actual responsivity (0.3 A/W at 265 GHz) from the real cited paper (Lischke et al., Nature Photonics 2021 — JANUS's own reference [29]). This is worse than the paper's

original claim, likely because that figure assumed a higher effective transimpedance gain than is achievable at 100 GHz — TIA gain and bandwidth trade off directly against each other, a real device constraint.

Rejected Noise Floor Claims

Claim	Status	Why Rejected
“Noise floor below -40 dBm at 100GHz”	Rejected	Likely confusion between noise spectral density (dBm/Hz) and total integrated noise power (dBm), or between intrinsic bandwidth capability and actual operating noise floor.
~-46 dBm from dark-current shot noise alone	Rejected	Arithmetically correct but accounts for only one non-dominant noise term; its own source material stated TIA thermal noise usually dominates — exactly the -17.2 dBm figure used here.
“265GHz gives a better noise floor than 100GHz”	Rejected	265 GHz is the detector's intrinsic response speed, not a noise number. Higher operating bandwidth always costs more noise, never less.

Real, unquantified mitigation: matched-filter / integrate-and-dump receiver design (standard optimal receiver theory) — estimated +2 to +4 dB improvement over the brick-wall estimate above. This is a real technique, not yet built into a simulation to confirm the exact number (see Part III Open Items).

2.3 Other Resolved Design Questions

3-Tile GST Stacking — Kept, Not Reduced

Initial concern: thermal crosstalk between three vertically stacked active PCM layers during simultaneous writes — the least literature-supported risk in the design. Resolved via write-scheduling, not a materials or geometry change: since JANUS uses a “set once, run many times” model rather than continuous active switching, the fix is a controller-level rule — never write vertically-adjacent tiers simultaneously, insert a cooldown delay between same-position writes across tiers. The original 3-tile design is retained.

3D Bonding Terminology — Clarified

Term	Definition
Wafer-to-Wafer (W2W)	Both sides full, undiced wafers at bond time
Die-to-Wafer (D2W)	One side pre-diced, placed on an intact wafer
Die-to-Die (D2D)	JANUS's actual regime — both sides pre-diced

Laser Count / Splitting Architecture — Finalized

Signal chain: seed laser → passive split → Si₃N₄ ring cascade amplification (§2 above) → further passive split → detection. Earlier proposals (large single-stage splits with no amplification, or amplification placed after a wasteful large split) were superseded once real energy-conservation and saturation checks were run against the actual signal chain.

3. Foundry & Supply Chain — Final State

3.1 Si₃N₄ Photonic Backbone — LIGENTEC (Switzerland)

Item	Detail
Process	AN800, LPCVD Si ₃ N ₄ , C-band optimized
Current spec (2026)	Loss < 0.5 dB/m · Q > 2×10 ⁷ · automotive-qualified 200mm line
Access	Direct (ligentec.com) or via VLC Photonics (Spain) — official PDK co-developer, offers design assistance
Europactice access	Not available — confirmed absent from Europactice's live 2026 photonics partner list
Pricing (~\$350/mm ²)	Unconfirmed — get a direct quote before budgeting
GST/PCM integration	Not offered — confirmed absent from LIGENTEC LABS' advanced-integration menu

3.2 CMOS Layer — SCL Mohali / Dholera (India), via C2S

Item	Detail
SCL Mohali live process	180nm — not 90nm as originally assumed
SCL modernization	Targeting 65/40/28nm; tender/land-dispute stage as of Aug 2025 — 2027 uncertain
Dholera (Tata, new fab)	Targets 28–90nm; first silicon aimed Dec 2026 — more realistic for a 2027 90nm need, but early-stage
Access route	C2S (Chips to Startup, India Semiconductor Mission) — subsidizes MPW cost, user-confirmed eligible
Typical node via C2S	65nm most common at prototype stage; routes internationally when domestic node unavailable

3.3 GST / PCM Switches — LioniX International (Netherlands), via Europactice

Item	Detail
Confirmed role	Heater structure on SiN waveguide (TriPleX) — not the GST material itself
GST deposition	Separate post-process step, rented cleanroom, hired professionals — heterogeneous/BEOL addition
“30% discount” claim	Unconfirmed — get in writing before relying on it
IMECAS (China) route	Rejected — real published capability, deprioritized for export-control / IP custody concerns, not fully reviewed

Important corrected distinction: IMECAS (Institute of Microelectronics, Chinese Academy of Sciences) and imec (Belgian institute, used for bonding below) are different organizations that share a similar acronym — this was a real source of confusion earlier in the process.

3.4 Die-to-Die Bonding — imec (Belgium)

Record	Pitch / Density	Regime	Applies to JANUS?
imec + EVG, 2026	200nm / 25,000,000/mm ²	Wafer-to-Wafer (W2W)	No — wrong regime
imec, 2024	2μm / ~250,000/mm ²	Die-to-Wafer (D2W)	Optimistic ceiling only

Record	Pitch / Density	Regime	Applies to JANUS?
True Die-to-Die (D2D)	No published record found	JANUS's actual regime	Unverified — top priority to confirm

The 900,000/mm² Cu-Cu dummy fill density proposed for thermal/mechanical bonding is achievable for W2W, but JANUS's real regime (D2D) realistically caps closer to the ~250,000/mm² D2W figure above — itself unconfirmed for true die-to-die.

3.5 Rejected Broker / Vendor Assumptions

- **Europractice as LIGENTEC's broker** — factually incorrect; corrected to VLC Photonics.
- **“AMD” for heater attachment** — never resolved; likely a placeholder for an unrecalled vendor name. Open item.
- **C2S naming collision** — a separate French CNFM service shares this acronym; the India Semiconductor Mission program is the one actually in use here.

PART II — Design History & Verification Trail

This part preserves the complete reasoning chain behind Part I: the original June 2026 physics-level error audit (Section A), the full GaP amplifier investigation that GaP followed before being set aside for foundry reasons (Section B), and the Si₃N₄ paths that were tested and rejected before arriving at the locked cascade design (Section C).

A. Physics-Level Error Analysis & Corrections (June 2026)

Prepared from cross-document analysis of the original JANUS patent specification (Janus.pdf) and an independent stress-test session. Five verified errors were originally identified across the timing model, GaP microring spectral matching, optical power budget, laser source specification, and CMOS backend design; a sixth was introduced and resolved by the 100 GHz upgrade itself. None of the errors affect the core architectural innovations — One-Hot spatial encoding, PCM Beneš routing, CRT fault visibility, or Z-axis thermal partitioning — which remain physically valid.

Error Inventory at a Glance

Error	Section	Nature	Impact	Status
E1: Timing model	IV.J.1 Eq.247–250	Serial sum applied to pipelined system	33→100 GHz, 3× throughput gain	Fixed: wave-pipeline model
E2: GaP Q-factor	III.D.4	$Q > 10,000$ with 1.5ps pulses — 15× spectral mismatch	Power loss, broken timing	Fixed: $Q \approx 2,000$, 5ps pulses
E3: Detector power	III.B.5 narrative	−7.5dBm stated vs +2.5dBm actual	Margin understated by 10dB	Fixed: narrative correction
E4: Laser power	III.E.3, Table XVII	1.6mW avg vs 0.032mW correct; Table XVII ×50 too large	Laser budget overstated 50×	Fixed: 0.44W datacenter
E5: MLL WPE	III.E, Table XVII	60% WPE MLL assumed; demonstrated max 2–10%	Corrupts full power budget	Fixed: CW+TFLN+compressor, 40% WPE
E6: CMOS backend	IV.J.1.a (new)	Single pipeline cannot run at 100 GHz	Undefined bottleneck at 100GHz	Fixed: 21 parallel pipelines at 4.76GHz

Error 1 — Timing Model (Most Fundamental)

Section IV.J.1, Equations 247–250. The paper's claim: $T_{\text{cycle}} = t_{\text{EAM}} + t_{\text{PD}} + t_{\text{wire}} + t_{\text{guard}} = 10 + 0.60 + 1.33 + 18.6 \approx 30.5 \text{ ps} \rightarrow 33 \text{ GHz}$.

Fatal logic error: serial summation applied to a wave-pipelined system. This equation treats every latency term as a serial throughput bottleneck, which is correct only for a single-stage pipeline with no overlap. JANUS is, by construction, deeply wave-pipelined — each component processes a different pulse simultaneously. Summing their latencies gives end-to-end single-computation latency, not the throughput interval between successive pulses.

Component-by-component verdict

Component	Role	Value	Throughput Bottleneck?
EAM modulation	Modulates each new pulse	10 ps → 100 GHz	YES — limits throughput
Ge-fin detector	Recovers between pulses	0.60 ps → 265 GHz	NO — not limiting
Wire propagation	Latency only, pipelined	1.33 ps (latency)	NO — multiple pulses coexist

Component	Role	Value	Throughput Bottleneck?
Beneš 15-stage routing	750 ps total latency, pipelined	750 ps (latency only)	NO — wave-pipelined
PCM settling	Once per 0.1ms epoch	~0.1 ms (amortised)	NO — not per-cycle
CMOS CRT adder-tree	210 ps latency, pipelined	210 ps (latency only)	NO — runs in parallel

The most damaging term was $t_{\text{guard}} = 18.6 \text{ ps}$ for PCM routing state stabilization. PCM switches are non-volatile; once programmed for an inference epoch (~0.1 ms), they are passive glass and require settling only after a weight-loading event. Adding 18.6 ps of settling overhead to every 30.5 ps clock cycle is physically nonsensical — this single spurious term accounted for 61% of the stated cycle time.

Corrected throughput-limiting equation: $f_{\text{clk}} = 1 / \max(t_{\text{EAM}}, t_{\text{detector_recovery}}, t_{\text{CMOS_latch_setup}}) = 1 / 10 \text{ ps} = 100 \text{ GHz}$ (EAM-limited). End-to-end single-computation latency (not clock period): $T_{\text{latency}} = 750 \text{ ps} + 210 \text{ ps} = 960 \text{ ps}$ (10 pipeline stages at 100 GHz, vs. 31 stages implied at the original 33 GHz claim).

Note on the 33 GHz figure: it may coincidentally approximate the CMOS backend's sustainable throughput ceiling for a single pipeline (4–6 GHz at TSMC N3), which must be resolved via parallelism (Error 6). The stated derivation is wrong regardless; the optical system itself supports 100 GHz.

Throughput Impact

Configuration	Original (33GHz, 64-bit)	Corrected (100GHz, 64-bit)	Multiplier
Mini (32 tiles)	57.6 TMAC/s	~174.5 TMAC/s	~3.0×
Edge (32 tiles)	229.8 TMAC/s	~696.5 TMAC/s	~3.0×
Datacenter (32 tiles)	919.2 TMAC/s	~2,785 TMAC/s	~3.0×

Note: all throughput figures represent exact integer 64-bit operations. The 3× gain is purely from correcting the timing model; no architectural change is required.

Error 2 — GaP Microring Spectral Mismatch (Original 33 GHz Analysis)

Section III.D.4 originally specified GaP microring resonators with $Q > 10,000$ (radius $\approx 20\mu\text{m}$) for phase-sensitive amplification, with 1.5 ps signal pulses generated by a mode-locked laser at 33 GHz.

Spectral mismatch: pulse ~15× wider than cavity passband. Cavity bandwidth at $Q=10,000$, $\lambda=1550\text{nm}$ ($f_0\approx193 \text{ THz}$): $\Delta f_{\text{cavity}} = f_0/Q = 193 \text{ THz} / 10,000 = 19.3 \text{ GHz}$. Transform-limited Gaussian pulse bandwidth at 1.5ps: $\Delta f_{\text{pulse}} = 0.44/\tau = 0.44/1.5\text{ps} \approx 293 \text{ GHz}$. The ring acts as a brutal spectral filter, rejecting the overwhelming majority of the pulse's energy and temporally stretching the transmitted remnant, breaking both power budget and timing closure simultaneously.

Fix applied at the time: pulse width 1.5ps → 5ps; GaP ring $Q \ 10,000+ \rightarrow \sim 2,000$. Verification: $\Delta f_{\text{pulse}}(5\text{ps}) = 0.44/5\text{ps} \approx 88 \text{ GHz}$; $\Delta f_{\text{cavity}}(Q=2000) = 193 \text{ THz}/2,000 \approx 96.5 \text{ GHz}$ — the 96.5 GHz cavity accommodates the 88 GHz pulse with ~9% margin.

Parameter	Original (wrong)	Corrected (2000-Q fix)
Pulse width	1.5 ps	5 ps
Pulse bandwidth (0.44/τ)	~293 GHz	~88 GHz
GaP ring Q-factor	>10,000	~2,000
Cavity passband (f_0/Q)	~19.3 GHz	~96.5 GHz
Spectral match ratio	15× mismatch (broken)	≈1.1× (matched with margin)
Pump external power / site	~1.3 μW	~6.5 μW (negligible increase)

Parameter	Original (wrong)	Corrected (2000-Q fix)
Superseded: this Q≈2,000 GaP-in-signal-path fix was itself a temporary patch. The full GaP investigation (Part II-B) later found this Q value could deliver at most single-digit dB of gain — nowhere near the 30 dB needed — which drove the architectural change of moving amplification to the CW carrier pre-pulse-formation, and eventually the switch to Si ₃ N ₄ for foundry reasons (Part I §2).		

Error 3 — Detector Signal Power Inconsistency

Section III.B.5 explicitly stated: “P_{signal} ≈ -7.5 dBm” at the Ge-fin detector, leading to a claimed detection margin of (-7.5) - (-31.76) = +24.26 dB.

Tracing the optical chain defined by the paper itself: Laser (peak, external) → 1:64 passive split -18.06 dB → GaP PSA front-end +30.00 dB → Beneš 15-stage routing -7.50 dB. Net offset from laser: +4.44 dB. If laser input to splitter ≈ -2 dBm (consistent with the budget), the detector receives: -2 + 4.44 = +2.5 dBm.

Narrative error: the routing loss magnitude (-7.5 dB) was used directly as the detector signal level, ignoring the +30 dB PSA gain that precedes it. The correct detector power is +2.5 dBm, not -7.5 dBm. Table VIII (‘Net Optical Drift ≈ 0 dB’) is internally consistent with +2.5 dBm — the table is correct; the narrative text is wrong.

Corrected detection margin: detector signal +2.5 dBm; noise floor after 33 GHz low-pass filter -31.76 dBm; margin = (+2.5) - (-31.76) = +34.26 dB — 10 dB larger than stated, increasing robustness to fabrication loss variations, but requiring correction in the narrative before external review.

Error 4 — Laser Power Budget Arithmetic

Section III.E.3 (p.16) stated each laser delivers ~1.6 mW time-averaged power (-20 dBm peak, 5% duty cycle, split loss accounted). Table XVII then used 1.6 mW × 8,192 lasers / 60% WPE = 13.33 W for the datacenter laser budget.

Factor-of-3,200 arithmetic error: at -20 dBm peak = 0.01 mW and 5% duty cycle, time-average = 0.01 mW × 0.05 = 0.0005 mW (0.5 μW) — the paper's claimed 1.6 mW is 3,200× larger. Since Table XVII's 13.33 W laser budget derives from this figure, the entire laser power row was ~50× too large.

Correct derivation (working forward from the detector requirement): Detector target +2.5 dBm (corrected, Error 3); + Beneš routing loss 7.5 dB; - GaP PSA gain 30 dB (undoing gain); + 1:64 passive split 18.06 dB → required laser peak = 2.5 + 7.5 - 30 + 18.06 ≈ -1.94 dBm ≈ 0.64 mW peak. Time-averaged laser power = 0.64 mW × 0.05 (duty cycle) = 0.032 mW per laser.

Total datacenter laser power: 0.032 mW × 8,192 lasers / 0.40 WPE (CW+TFLN) ≈ 0.66 W. Using the original 60% WPE MLL assumption instead: 0.032 mW × 8,192 / 0.60 ≈ 0.44 W.

Corrected Laser Power Budget
Per-laser time-averaged optical power: 0.032 mW (not 1.6 mW). Total datacenter laser electrical power: ~0.44-0.66 W (not 13.33 W) — a reduction of ~20-30×. Total system power for the Datacenter configuration drops from 120-140 W to approximately 107-127 W once this correction is applied (other rows unaffected).

Error 5 — Mode-Locked Laser Wall-Plug Efficiency

Table XVII listed the laser source as ‘Laser Source (60% WPE)’, applied to a mode-locked pulsed laser (MLL) running at 33 GHz repetition rate with 1.5 ps pulses.

60% WPE is demonstrated only for CW lasers, not MLLs. The cited 60% figure (Ren et al., 2020, J. Semiconductors) is for a continuous-wave 808nm laser diode optimized for DC operation. State-of-the-art MLLs at 10-100 GHz repetition rates

demonstrate only 2–10% WPE (passive MLLs 1–5%, active MLLs with RF drive 3–8%). 60% WPE is physically unreachable for any pulsed GHz laser today; applying the CW figure to an MLL would have understated actual MLL power consumption by 6–30×.

The Industry-Standard Fix: CW + TFLN MZM + Pulse Compressor

This is the canonical solution for high-repetition-rate electro-optic pulse generation, used in all production 100G/400G optical transceivers:

- **Step 1 — CW InP-on-Si laser:** 30–40% WPE demonstrated, wavelength-stable, single-mode. Replaces the MLL entirely.
- **Step 2 — TFLN Mach-Zehnder Modulator (MZM):** bandwidth >100 GHz demonstrated. Pockels effect (purely capacitive), $V_{\pi} < 1V$, insertion loss ~2.5 dB, sub-fJ/bit energy. Driven with a 100 GHz sinusoidal RF signal.
- **Step 3 — Chirped Bragg grating pulse compressor:** passive, ~1.5 dB insertion loss. Compresses the sinusoidal modulation into ~5 ps pulses. LiTaO₃ is already in the JANUS material stack (Table XVII Pockels ring routers), so no new material platform is required.

Combined insertion loss budget: TFLN MZM ~2.5 dB + chirped Bragg grating ~1.5 dB = total ~4.0 dB, easily offset by the 6× WPE improvement over an MLL (40% CW vs. ~7% MLL average). Net effect: lower laser source power consumption with a more manufacturable, commercially available source.

Corrected Laser Architecture

Source: CW InP-on-Si laser at 1550nm, 30–40% WPE (design point 40%). Modulator: TFLN MZM at 100 GHz, $V_{\pi}<1V$, IL=2.5dB. Pulse shaping: chirped Bragg grating compressor, IL=1.5dB → 5ps output pulses. Combined effective system WPE: ~35% after insertion loss. Table XVII laser row: replace ‘MLL 60% WPE’ with ‘CW+TFLN+CBG 35–40% WPE’.

References for the CW+TFLN Solution

- TFLN modulator >100 GHz bandwidth: Wang et al. (2018), Nature, doi:10.1038/s41586-018-0551-y
- TFLN MZM with $V_{\pi} < 1V$: He et al. (2019), Nature Photonics, doi:10.1038/s41566-019-0378-6
- InP-on-Si CW laser 40% WPE review: Coldren et al., IEEE J. Sel. Top. Quantum Electron.

Error 6 — CMOS Backend Bottleneck at 100 GHz

A new issue introduced by the 100 GHz upgrade itself: once the optical pipeline is correctly clocked at 100 GHz, optical results arrive every 10 ps. The CRT reconstruction pipeline has ~210 ps total latency. A single serial CMOS pipeline at TSMC N3 cannot close timing at 100 GHz — complex adder trees operate at only 4–6 GHz on N3 even under maximum voltage stress.

Solution — time-interleaved parallel pipelines: the pipeline does not need to run at 100 GHz, it only needs to produce results at a 100 GHz rate. $N_{\text{pipelines}} = t_{\text{CRT}} / t_{\text{optical}} = 210 \text{ ps} / 10 \text{ ps} = 21$ parallel pipelines. Each pipeline runs at $f_{\text{CMOS}} = 100 \text{ GHz} / 21 \approx 4.76 \text{ GHz}$ — comfortable at TSMC N3.

Why This Is Standard Industry Practice

Application	Technique	Parallelism Factor	Reference
High-speed ADCs (>10 GSp/s)	Time-interleaved ADC cores	4–16×	TI ADC12DJ5200RF, ADI AD9213
100G/400G Ethernet SerDes	1:N DMUX after TIA	4–8×	400G QSFP-DD (Inphi, Marvell, Broadcom)
HBM memory interface	Wide parallel buses at lower f	128–1024×	HBM2E, HBM3 (JEDEC)

Application	Technique	Parallelism Factor	Reference
Apple M-series die-to-die	Parallel slow lanes	Many	Apple Silicon (WWDC 2020)
DSP polyphase filter banks	Polyphase decomposition	$N\times$	Vaidyanathan, Multirate Systems and Filter Banks

Power advantage: dynamic CMOS power $P = \alpha \cdot C \cdot V^2 \cdot f$. At TSMC N3, 4.76 GHz operation allows $V_{DD} \approx 0.65V$ (comfortable, near-threshold), whereas a hypothetical 100 GHz pipeline would require $V_{DD} > 1.0V$ (impossible for complex logic). Power ratio (single fast vs. 21 parallel slow) = $(1.0^2 \times 100) / (21 \times 0.65^2 \times 4.76) = 100 / 42.2 \approx 2.37\times$ — the parallel approach uses roughly less than half the dynamic power of a hypothetical single fast pipeline, at identical throughput.

JANUS Integration Architecture

Ge-fin detector (265 GHz bandwidth) → TIA (already in Table XVII I/O Interface) → 1:21 high-speed electronic DMUX (SiGe BiCMOS or InP HBT, demonstrated at 100 GHz) → 21 parallel CRT reconstruction pipelines at ~4.76 GHz (TSMC N3, easy timing closure) → round-robin output MUX back to a 100 GHz result stream.

Sequencing: CRT reconstruction requires residues from all k optical tiles for the same computation to arrive at the same pipeline instance. Results are tagged with a computation ID by the CMOS encoder front-end; the DMUX routes by ID mod 21. Since all k residue tiles for one computation are clocked synchronously (shared optical master clock), they arrive within the same 10 ps window and are routed together — no out-of-order reconstruction problem arises.

Area: 21 CRT engines at TSMC N3 (~170 MTr/mm²) fit comfortably in the existing CMOS footprint. 1:21 DMUX at 100 GHz input uses a SiGe BiCMOS front-end, identical to 100G/400G transceiver chipsets.

B. GaP Parametric Amplifier Investigation (July 2026)

Status note: this entire investigation converged on a locked GaP specification (§B.9 below). That specification is physically sound but requires a loaded Q (~290,000–300,000) that is not currently obtainable via any accessible MPW run — no foundry in the current supply chain (Part I §3) offers GaP-on-insulator rings at that Q. The design-of-record was therefore moved to Si₃N₄ (Part I §2). GaP remains the preferred upgrade path if a suitable MPW becomes available.

This section consolidates a full engineering trace of the JANUS optical amplification stage as originally pursued: why a discrete SOA baseline was rejected, why GaP was selected as the nonlinear gain medium over five competing materials, the original design flaw that made the initial ‘5 microwatt pump for 30dB gain’ target physically unreachable, every architecture explored to close that gap, and the final locked GaP specification with its power budget.

B.1 Background — Why Amplification Is Needed

JANUS routes optical signals through a 15-stage PCM Beneš fabric to implement reconfigurable spatial matrix computation. Before entering this fabric, each signal is passively split 1:64 to fan out across the computation array. A passive 1:64 splitter introduces an unavoidable insertion loss of $\text{Loss(dB)} = 10 \cdot \log_{10}(1/64) \approx -18.1 \text{ dB}$, which must be recovered somewhere in the optical chain before the signal reaches the Ge-fin photodetector. The original specification called for a per-site parametric amplifier delivering 30 dB of gain — enough to cover the split loss with ~12 dB of margin.

B.2 The SOA Baseline and Why It Was Rejected

The first candidate evaluated for the amplification role was a conventional discrete Semiconductor Optical Amplifier (SOA), the incumbent telecom technology.

SOA class	Electrical power draw	Notes
High-power discrete (BOA1554P-class)	~1.8 W per device	I≈1200mA, V≈1.5V; 4-SOA module draws 7.2W
Low-power variant (10–15 dB gain)	~50 mW per device	Lower bias current, reduced max gain
AWGR photonic-accelerator integrated SOA	42–84 mW per device	Scales with data rate and gain configuration

At JANUS’s scale of roughly 600,000 amplification sites (pre-consolidation), even the lowest-power SOA class evaluated (50mW) would require $600,000 \times 50 \text{ mW} = 30,000 \text{ W}$ (30 kW) — categorically incompatible with a chip-scale accelerator power budget. SOAs also introduce electrical-to-optical conversion inefficiency, amplified spontaneous emission (ASE) noise, and gain-recovery times on the order of hundreds of picoseconds (carrier recombination-limited), motivating a move to an all-optical, cavity-enhanced parametric gain mechanism where gain is instantaneous (femtosecond-scale, $\chi^{(2)}/\chi^{(3)}$ electronic-cloud polarization response) and driven optically rather than electrically.

B.3 Ring Resonator Material Screening

The nonlinear gain medium was selected against three simultaneous criteria: (1) Pockels/EO performance at or above TFLN, (2) strong Kerr ($\chi^{(3)}$) nonlinearity for parametric gain, and (3) two-photon-absorption-free (TPA-free) operation at the signal wavelength. Six candidate materials were evaluated.

Material	Verdict
TFLN (LiNbO ₃)	Strong EO (Pockels) performance; used elsewhere in JANUS (MZM, ring routers) but weaker intrinsic Kerr gain per unit pump than GaP
AlN	Excellent CTE match to GaP and native $\chi^{(2)}$; Pockels coefficient weak (~1–1.5 pm/V pure, ~3–10 pm/V Sc-doped) — later relevant as a high-Q host (§B.8)

Material	Verdict
BaTiO ₃	Recommended in a separate (non-JANUS) photonics screening for its coupled double-ring bistability profile; not carried into the JANUS PSA decision
GaP	Selected — see §B.3.1
Si	TPA-active at telecom wavelengths under high intensity; ruled out early
AlGaAs	Strong nonlinearity but narrower TPA-free window than GaP; not pursued further

B.3.1 Why GaP

- Kerr index $n_2 \approx 1.2 \times 10^{-17} \text{ m}^2/\text{W}$ — among the highest of CMOS/foundry-compatible integrated platforms, roughly 3× stronger than the GeSbS chalcogenide alternative (§B.8).
- TPA-free for photon energies below its indirect bandgap (2.26 eV) when combined two-photon energy stays under that threshold — satisfied at 1550nm ($2 \times 0.80 \text{ eV} = 1.60 \text{ eV}$, safely below gap) but not at 775–800nm ($2 \times 1.55\text{--}1.60 \text{ eV} = 3.10\text{--}3.20 \text{ eV}$, well above gap). This becomes decisive in §B.7.
- Real, demonstrated integration on silicon-compatible substrates (GaP-on-insulator), with published ring resonator Q factors and second-harmonic-generation conversion efficiency data available as calibration anchors (Logan et al.; Wilson et al.).
- Second-order ($\chi^{(2)}$) nonlinearity also present ($d_{\text{eff}} \approx 70 \text{ pm/V}$), enabling an alternative pump-degenerate architecture explored and ultimately rejected in §B.7.

B.4 The Original Design and Its Flaw

The original JANUS specification called for a GaP microring with $Q > 10,000$ for phase-sensitive amplification, later corrected (Part II-A, Error 2) to $Q \approx 2,000$ with 5ps pulses, to spectrally match a 100 GHz pulse train. Independently, a design target of ~5 microwatts of external CW pump power producing 30dB of parametric gain was carried forward at this Q. This section documents how that target was tested and where it failed.

B.4.1 The bandwidth-matching constraint that set $Q \approx 2,000\text{--}2,750$

A transform-limited Gaussian pulse of duration τ has spectral bandwidth $\Delta f_{\text{pulse}} \approx 0.44/\tau$. For the ring cavity to pass such a pulse without distorting or stretching it, its own passband must be at least this wide: $\Delta f_{\text{cavity}} = f_0/Q \geq \Delta f_{\text{pulse}} = 0.44/\tau$.

Pulse/clock rate	Q_max (bandwidth-limited)
20 GHz	~11,000
33 GHz	~6,700
50 GHz	~4,400
80 GHz	~2,750
100 GHz	~2,200

This confirmed the earlier $Q \approx 2,000$ correction was internally consistent for 80–100 GHz operation — but it also set a hard ceiling that could not be raised without dropping the pulse rate.

B.4.2 Buildup at this Q is far too small for 30 dB

Resonant power buildup in a critically-coupled ring is set by the cavity finesse: Finesse $F = Q/m$, where $m = n_g \cdot L/\lambda_0$ (mode number); Buildup $B \approx F/\pi$. For a 20μm radius GaP ring ($n_g \approx 3.3$, $\lambda_0 = 1550 \text{ nm}$), $m \approx 267$, giving at most ~4–11 dB of buildup across the Q range the bandwidth constraint allows — nowhere close to the ~30 dB the design target required from resonant enhancement alone.

B.4.3 Radius does not close the gap — tested exhaustively

A natural hypothesis was that shrinking the ring radius could recover the missing gain, since smaller rings have fewer modes (m) and therefore higher finesse for fixed Q . This was tested down to $R=1\text{--}2\mu\text{m}$. The result: total effective interaction length inside the cavity, $L_{\text{eff}} = N_{\text{roundtrips}} \times L = Q \cdot \lambda_o / (2\pi \cdot n_g)$, depends only on Q , not on radius — the radius-dependence in the number of round trips exactly cancels the radius-dependence in the round-trip length. Smaller radius does increase peak circulating buildup (roughly 4dB gained from $5\mu\text{m} \rightarrow 2\mu\text{m}$), but total achievable parametric gain still fell far short of 30dB at the bandwidth-limited Q , and sub- $3\mu\text{m}$ radii approach fabrication limits where bend-radiation loss would likely erode Q before any gain benefit is realized.

Conclusion of §B.4: at the Q the 80–100 GHz pulse rate forces ($\sim 2,000\text{--}2,750$), no realistic combination of radius, coupling length, or pump power within a single ring reaches 30dB of parametric gain from microwatt-level pump. First-principles calculation using GaP's known Kerr index put the actual pump requirement at approximately 5–10 W for 15–30dB at this Q — roughly six orders of magnitude above the 5 microwatt target.

B.5 Root Cause: The Q-vs-Bandwidth Wall

The underlying problem is that a single ring resonator is being asked to do two incompatible jobs simultaneously: pass a wideband pulsed data signal (requiring low Q) and build up enough circulating intensity for strong parametric gain (requiring high Q). These pull in opposite directions on the same parameter.

B.5.1 Quantitative pump power at the bandwidth-limited Q

Using the standard undepleted-pump parametric gain formula for four-wave mixing, $G(\text{dB}) = 10 \cdot \log_{10}(\cosh^2(\gamma \cdot P_{\text{circ}} \cdot L_{\text{eff}}))$, $\gamma = 2\pi \cdot n_2 / (\lambda \cdot A_{\text{eff}})$, with GaP's $n_2 \approx 1.2 \times 10^{-17} \text{ m}^2/\text{W}$ and a representative mode area $A_{\text{eff}} \approx 0.2\mu\text{m}^2$, the pump power required at $R=5\mu\text{m}$, $Q \approx 2,200$ was calculated directly:

Target gain	Required external pump (single-ring, bandwidth-limited Q)
10 dB	$\sim 4.3 \text{ W}$
15 dB	$\sim 5.8 \text{ W}$
30 dB	$\sim 9.9 \text{ W}$

This result was cross-checked using an alternative, looser effective-length approximation sometimes used informally ($L_{\text{eff}} \approx Q \cdot 2\pi R$, treating Q as if it equaled the number of physical round trips directly). Even this generous approximation — which overstates the true effective length by a factor of $2\pi \cdot m$ ($\approx 420\times$ at $R=5\mu\text{m}$) — still requires pump power in the low tens of milliwatts, not microwatts, confirming the gap was not merely a modeling artifact.

B.5.2 The resolution: decouple the signal path from the pump-buildup path

The wall is only present because the ring was assumed to carry the already-pulsed 80–100 GHz signal. If amplification instead acts on the CW optical carrier before pulse formation — i.e., before the TFLN MZM and chirped Bragg grating compressor stage that creates the 5ps pulses — the ring never needs to pass wideband pulsed content. A CW tone has no bandwidth requirement beyond sitting on resonance, so Q is free to rise to whatever value the fabrication process actually supports, with no bandwidth penalty. This single re-ordering — amplify-then-pulse rather than pulse-then-amplify — is what made the rest of the GaP specification (and, later, the Si_3N_4 cascade of Part I §2) possible.

B.6 Geometry Study: Radius, Buildup, and Effective Length

With the bandwidth constraint removed, Q is no longer capped at $\sim 2,750$. The remaining design question is what combination of radius and Q is both physically realistic and pump-efficient.

B.6.1 Buildup vs. radius, fixed Q

For a fixed Q , buildup scales as $1/m$ — inversely with radius (fewer modes in a smaller ring raises finesse). Confirmed numerically:

Radius	Mode number (m), Q=2,000	Buildup (dB)
20 μm	268	7.7
10 μm	134	10.8
5 μm	67	13.8
2 μm	27	17.9 (fabrication risk)

Because total effective interaction length is radius-independent (§B.4.3), this buildup gain alone was insufficient to reach 30dB at bandwidth-limited Q; it becomes relevant again once Q is freed from the bandwidth constraint.

B.6.2 Fabrication floor

Radii below approximately 3–5 μm risk bend-radiation loss dominating over material/scattering loss, degrading Q below the value assumed in any buildup calculation. R=5 μm was retained as the practical floor because it coincides with the radius of the real, demonstrated GaP-on-insulator ring resonator used as the Q calibration anchor for the final specification.

B.6.3 Final geometry sweep, single-wavelength, real/near-real Q

Once the amplifier was moved to the CW pre-pulse position and the wavelength question resolved to single-wavelength ~1550nm operation (§B.7), the following sweep was performed using the $\chi^{(3)}$ Kerr gain formula from §B.5.1:

Radius	Q	Pump for 21dB / site	Total, 8,192 sites
10 μm	290,000 (measured)	854 μW	7.0 W
10 μm	300,000 (near-term)	798 μW	6.5 W
5 μm	290,000 (measured)	427 μW	3.5 W
5 μm	300,000 (near-term)	399 μW	3.3 W

Halving the radius from 10 μm to 5 μm at fixed Q roughly halves the required pump power, consistent with the 1/m buildup scaling — with total interaction length held constant by Q, the buildup advantage translates directly into a pump-power advantage rather than being cancelled out. R=5 μm , Q≈290,000–300,000 was carried forward as the final GaP geometry.

B.7 Wavelength Architecture: Single-Wavelength vs. 775/1550 nm

B.7.1 Option 1 — $\chi^{(2)}$ degenerate OPA (775nm pump / 1550nm signal)

GaP's second-order nonlinearity ($d_{\text{eff}} \approx 70 \text{ pm/V}$) supports parametric down-conversion from a 775nm pump to a degenerate 1550nm signal/idler pair. Attractive because 775–800nm diode lasers can reach WPE above 60%, substantially higher than typical 1550nm telecom laser WPE (~30–40%). However: $2 \times E_{\text{photon}}(800\text{nm}) = 3.10 \text{ eV} > E_{\text{gap,indirect}}(\text{GaP}) = 2.26 \text{ eV}$. At 775–800nm, combined two-photon energy exceeds both GaP's indirect (2.26eV) and direct (2.78eV) bandgaps — TPA is strongly active. GaP only remains TPA-free above ~1,097nm. This reintroduces the exact loss mechanism that was one of the three original screening criteria for selecting GaP (§B.3).

This mechanism was quantified anyway for comparison, using the coupled-wave OPA gain formula (Boyd normalization): a pure GaP ring at its real demonstrated Q ceiling (290,000) required approximately 30.8 mW per site for 21dB gain — roughly 36× more pump power than the equivalent single-wavelength χ^3 design at the same Q and radius, because χ^2 gain at this Q only accumulates over ~345 recirculation passes versus the much longer effective interaction length available to the χ^3 path. Rejected on combined TPA-risk and higher-realized-pump-power grounds, despite the laser-efficiency advantage.

B.7.2 Option 2 — $\chi^{(3)}$ Kerr four-wave mixing, single-wavelength (~1550nm) — ADOPTED

Pump and signal both sit near 1550nm (non-degenerate FWM, small detuning between pump/signal/idler), keeping every photon below GaP's TPA threshold and preserving JANUS's core single-wavelength invariant (which independently exists

to suppress FWM/XPM cross-talk in the routing fabric — Part II-A). This is the mechanism used for all final GaP power-budget figures (§B.9).

B.7.3 Ge-fin photodetector compatibility check

Because the 775/800nm option was under consideration, the downstream Ge-fin photodetector (265 GHz bandwidth, Lischke et al. 2021) was checked for compatibility at that wavelength. Germanium's absorption coefficient is strong from the visible through 1550nm and only begins dropping sharply above ~1550nm — meaning 775–800nm sits well within Ge's high-absorption band, and a thinner absorption layer (feasible at this wavelength) would if anything favor higher bandwidth via reduced carrier transit time. No literature demonstration of the specific 265 GHz design re-optimized for 800nm was found, but no physical obstacle was identified either. This check became moot once the 775nm pump path was rejected on TPA grounds, but is retained as it may be relevant if a χ^2 path is revisited with a different, TPA-safe host material.

B.8 Hybrid High-Q Host Exploration

Before settling on the pure-GaP, real-Q-anchored design, a parallel line of investigation explored whether a separate ultra-high-Q host material (evanescently coupled to a thin GaP nonlinear layer) could beat pure GaP's own Q ceiling. Retained here because it surfaced a materially important methodological finding, and because it may become the better path if GaP's own fabrication ceiling does not improve.

B.8.1 Candidate host materials

Material	Best demonstrated Q (2025–2026 lit.)	Notes
Si ₃ N ₄	~1.1×10 ⁷ (subtractive, foundry); >2×10 ⁸ (absorption-limited, ultralow-loss wafer-scale)	Most mature ultra-high-Q integrated platform
TFLN / LiNbO ₃	1.4×10 ⁷ (vertical evanescent coupling to SiN, 2025); 4×10 ⁷ (all-LN microdisk, PLACE)	Already present elsewhere in JANUS material stack
AlN (single-crystal, sapphire)	2.8×10 ⁶ (2025)	CTE-matched to GaP; reintroduces sapphire CTE mismatch
AlN (racetrack, optimized)	5.4×10 ⁶ (Jan 2026)	Foundry-compatible, no sapphire substrate
GeSbS (chalcogenide)	2–5×10 ⁶ (R≈100μm)	Real demonstrated 31.5dB gain at 8.5mW pump — but this is the material itself acting as the nonlinear medium, not a GaP host (§B.8.3)

B.8.2 The citation-mismatch finding

Important correction made during this investigation: an early version of the internal design reasoning cited “Xiao et al., OFC 2024, Th1D.4 — GaP integrated OPA, 31.5dB gain at 8.5mW pump” as a calibration anchor. Checking this citation directly found it does not describe a GaP device at all — Th1D.4 (Xiao, Xia, Luo, Li, Zhang) demonstrates a GeSbS microresonator. The real demonstrated GaP parametric amplifier in the 2024–2025 literature (Kuznetsov et al.) is an unrelated device type: a 5.55cm traveling-wave spiral waveguide with no resonant cavity, reaching 35dB gain via long propagation length and dispersion engineering rather than resonant buildup, at substantially higher (~100+mW) pump power. Neither cited paper is a valid GaP-microring calibration point, and this was corrected before further calculations relied on it.

B.8.3 GeSbS is a real high-Q, low-power result — but at the wrong Q for JANUS's original bandwidth constraint

The real GeSbS result (Q≈2×10⁶–5×10⁶, R≈100μm, 31.5dB at 8.5mW) was checked against the 80 GHz bandwidth requirement that originally constrained the signal-path ring. Its cavity bandwidth (39–97 MHz) is roughly 1,000× narrower

than the ~70–90 GHz an 80 GHz pulse train requires. This confirmed that no single-ring material choice resolves the bandwidth-vs-gain conflict — the constraint is architectural (§B.5.2), not a materials problem, which was the basis for moving amplification to the CW pre-pulse position.

B.8.4 Hybrid GaP-on-host evanescent coupling: quantified but unverified

With amplification moved to the CW carrier, a GaP-nonlinear-layer-on-high-Q-host architecture was quantified (TFLN host, $Q=1.4\times10^6$, GaP modal overlap 30–44% based on a comparable Si/III-V hybrid confinement demonstration): approximately 42–72μW pump per site for 30dB, or roughly 50μW at a 44% overlap figure. This 44% figure, however, was sourced from an unrelated device type (a Si/III-V gain waveguide, not a resonator, and not a GaP/TFLN pairing) — no published device has demonstrated high-Q and high-GaP-overlap simultaneously in one structure. This combination is flagged as aspirational, not validated (Part III).

A further sweep tested whether pushing this hybrid host to $Q\approx10^7$ (real Si_3N_4 /TFLN ceiling) could reach single-digit microwatt pump power. It does ($\approx1\text{--}2\mu\text{W}$ at $Q=10^7$, 30% overlap) — but the hybrid integration, bonding-interface loss, and dual-material CTE/lattice matching this requires were judged higher-risk than the pure-GaP path adopted below.

B.9 Final Locked GaP Specification

This was the design-of-record at the time this GaP investigation concluded. It has since been superseded by Si_3N_4 for foundry-availability reasons only (see status note at the top of Part II-B and Part I §1).

Parameter	Value
Nonlinear gain medium	Pure GaP microring (no hybrid host material)
Nonlinear mechanism	$\chi^{(3)}$ Kerr four-wave mixing, single-wavelength (~1550nm pump and signal)
Ring radius	5 μm
Loaded Q	$\approx290,000$ (real, demonstrated GaP-on-insulator ring) to 300,000 (near-term projection)
Target gain per amplifier	21.0 dB (exactly covers 1:64 split loss of ~18.1dB, ~2.9dB margin)
Amplifier position in datapath	Pre-pulse-formation (on CW carrier), pre-1:64-fanout
Number of amplifier instances	8,192 (one per laser source, matching datacenter laser count)
Pump power per site ($Q=290,000$)	$\approx427\text{ }\mu\text{W}$
Pump power per site ($Q=300,000$)	$\approx399\text{ }\mu\text{W}$
Total amplification-stage power ($Q=290,000$)	$\approx3.5\text{ W}$
Total amplification-stage power ($Q=300,000$)	$\approx3.3\text{ W}$
Share of total JANUS datacenter system power (~107–127W)	$\approx3\%$
TPA status at operating wavelength	TPA-free ($2\times0.80\text{eV} = 1.60\text{eV}$, below GaP's 2.26eV indirect gap)

B.9.1 Why this was the strongest-grounded GaP result in the design trace

Every other configuration considered (hybrid AlN/SiN host, χ^2 775nm pump, unverified $Q=3\times10^6$ pure GaP) relied on at least one unverified assumption — either an un-demonstrated Q, a mismatched literature citation, or a materials combination with no published device. This final specification uses only a real, measured GaP ring Q, a single wavelength consistent with GaP's own TPA-free window, and an architecture (pre-pulse CW amplification) whose timing behavior was independently checked rather than assumed.

B.9.2 Pipelining — Where Amplification Sits in the Datapath

Original position: post-fanout, one amplifier per output. The original architecture placed one amplifier at each of the ~600,000 post-1:64-split sites, after pulse formation — the configuration that ran into the Q-vs-bandwidth wall (§B.5) and, even after that wall was resolved, would still require 600,000 individually pump-delivered and individually EO-locked amplifier instances.

Corrected position: pre-fanout, on the CW carrier, before pulse formation. Two changes were made together: (1) amplification acts on the CW laser carrier, upstream of the TFLN MZM + chirped Bragg grating pulse-compression stage and upstream of the EAM data-modulation stage, removing the bandwidth constraint; (2) amplification is consolidated before the 1:64 fan-out, at the laser-count level (8,192 sites) rather than after it (600,000 sites). Since the 21dB gain target exists to cover the 1:64 split loss regardless of where in the chain it is applied, and per-amplifier pump power does not depend on how many downstream channels that one amplified signal eventually feeds, moving amplification upstream of the split reduces the amplifier count by the split ratio (64×, partially offset by the 8,192-vs-600,000 accounting) without any loss of coverage.

This reordering was checked for two potential problems, both resolved favorably: (a) Timing — because both pump and signal are CW at the point of amplification, the ring's photon lifetime ($\tau_{ph} = Q \cdot L / \omega_o$) manifests only as a one-time cavity-fill startup delay, not a per-symbol throughput bottleneck; all picosecond/GHz-scale data timing occurs downstream and is unaffected. (b) Optical intensity handling — consolidating amplification pre-fanout means each amplifier's output must supply 64× more downstream channels, raising absolute circulating/output optical power per site, which raises a legitimate follow-up question about pump depletion or TPA-edge proximity, flagged as an open item (Part III).

B.10 Alternatives Considered and Rejected (GaP Investigation)

Alternative	Why Rejected
Discrete SOA array	30kW at 600k sites even at the lowest-power SOA class evaluated; ASE noise; hundreds-of-ps gain recovery time
Single ring, $Q=2,000$ – $2,750$, post-fanout, post-pulse (original spec)	Bandwidth-limited buildup insufficient by ~6 orders of magnitude in pump power for 30dB gain
Sub- $5\mu\text{m}$ radius (2 – $3\mu\text{m}$) to boost buildup further	Diminishing returns (~4dB gain from $5 \rightarrow 2\mu\text{m}$) plus fabrication risk: bend-radiation loss likely erodes Q below assumed values at these radii
Cascaded 2–3 ring stages at moderate Q , post-fanout	Noise-cascade analysis (Friis formula) showed negligible noise penalty for cascading, but $\pm 2\text{nm}$ fabrication/etch tolerance versus sub-picometer cavity linewidths at high Q made independent dual-resonance phase-locking per ring a severe control-system cost (thousands of independently-locked resonance-trim loops at 600k scale)
$\chi^{(2)}$ 775nm pump / 1550nm signal, pure GaP	TPA strongly active at 775–800nm in GaP; also ~36× higher realized pump power than the χ^3 single-wavelength path at equal Q and radius
$\chi^{(2)}$ 775nm pump, hybrid AlN host ($Q=3.78 \times 10^6$)	TPA-safe (gain occurs in AlN region assumptions) but requires a second wavelength/laser type, additional lattice/CTE-matched hybrid integration, and still costs ~60W system-wide versus ~3.3–3.5W for the adopted GaP design
GeSbS as the nonlinear medium directly (real $Q=2$ – 5×10^6 demonstration)	Real, low-power result (31.5dB at 8.5mW) but only at cavity bandwidths (39–97MHz) incompatible with any architecture still requiring the ring to pass pulsed data; also ~3× weaker Kerr index than GaP
Hybrid TFLN/SiN host + GaP evanescent layer, $Q=10^6$ – 10^7	Physically promising (as low as ~1–2μW at $Q=10^7$) but requires an unverified combination of high host Q and high GaP modal overlap that has not been jointly demonstrated in one device; retained as a future path
Post-fanout amplifier placement	Timing-safe but power-inefficient: 64× more amplifier instances than the

Alternative	Why Rejected
(600k sites) with the corrected CW/pre-pulse timing fix	pre-fanout placement for no coverage benefit

B.11 Open Problems from the GaP Investigation

- Optical intensity / pump depletion check at the consolidated 8,192-site, 64x-downstream-channel amplifier output — not yet verified against GaP's TPA edge or gain-saturation behavior at the higher absolute power this consolidation implies.
- Non-degenerate FWM detuning: the exact pump-signal wavelength offset needed for practical signal/idler separation, balanced against the ring's phase-matching bandwidth at $Q \approx 290,000$ – $300,000$, has not yet been derived.
- Hybrid TFLN/SiN-host + GaP-evanescent-layer path remains open as a lower-power alternative if a real device demonstrating both high host Q and high GaP modal overlap is published, or if GaP's own fabrication ceiling does not improve beyond $\sim 300,000$ in the near term.
- Bend-loss characterization specific to GaP-on-insulator at $R=5\mu\text{m}$ has not been independently confirmed beyond the fact that this radius matches the real $Q=290,000$ demonstration's own geometry; a dedicated bend-loss-vs-radius dataset for GaP would tighten this.
- Pump routing and delivery to 8,192 sites (fiber array, on-chip splitting network, or shared laser bank) has not yet been designed.
- EO/thermal locking strategy for the final pure-GaP ring (post-fab trimming vs. active locking) has not been re-derived for this specific geometry; the AlN Pockels-based locking scheme explored in the hybrid-host thread does not directly transfer to pure GaP without re-evaluating GaP's own EO coefficient and electrode geometry at $R=5\mu\text{m}$.
- Foundry availability at the required Q ($\approx 290,000$ – $300,000$) via an accessible MPW run — currently unresolved; this is the reason the design-of-record moved to Si_3N_4 (Part I §1).

C. Si₃N₄ Amplifier — Rejected Paths

Before arriving at the locked Si₃N₄ cascade design in Part I §2 ($R=30\mu\text{m}$, $Q=3\times 10^6$, 11 rings), the following configurations were tested via the same coupled-mode-theory simulation methodology and rejected.

C.1 $Q = 20,000,000$ (the original target) — Rejected

The original target used LIGENTEC's highest advertised platform $Q (>2\times 10^7)$. Testing showed this Q is too high, not too low. A single ring at $Q=20\text{M}$, $R=10\mu\text{m}$ — at any pump power from threshold ($2.65\mu\text{W}$, confirmed by two independent calculation methods) up to $200\mu\text{W}$ — either self-oscillated or, at the real signal level from the actual seed/split chain, saturated to a fixed output ceiling around -32dBm , regardless of how seed power or split ratio were rearranged: $\sim 15\text{dB}$ short of the real noise floor, and unfixable by tuning.

C.2 $R=50\mu\text{m}$, $Q=1\times 10^6$ — 40.64dB, the highest gain found — Rejected

Technically passed the stability check at the exact point tested, but rejected for two reasons:

- **Cliff-edge behavior:** $2.0\text{mW}\rightarrow 3.0\text{mW}$ pump (+50%) jumped gain from 14.6dB to 34.1dB — a 19.3dB swing. Signature of operating beside a bifurcation, not a robust point.
- **5× higher power cost:** 55mW total pump vs. $11\text{--}16.5\text{mW}$ for the chosen design.

C.3 Single-ring 45dB claim from a real but misapplied paper — Rejected

arXiv 2209.05589 was cited claiming 45dB gain from $\sim 125\text{--}166\text{nW}$ input at $9\text{--}11\text{mW}$ pump. Checked directly against the source: the 45dB figure is simulated, not demonstrated (real result: 30dB at 9mW), and the paper states gain saturation onset at 100pW — the claimed input was $\sim 2000\times$ above that point. The real $30\text{dB}/9\text{mW}$ figure was instead used only as a literature sanity-check for the CMT model.

C.4 $Q = 7\times 10^7$ — Rejected

Exceeds even the best published academic Damascene record anywhere (EPFL, $\sim 3\times 10^7$) — not supported by any citation and not carried forward.

C.5 “0.5mW solves it via energy conservation” — Rejected

Energy conservation (pump $\geq$ energy added to signal) is necessary but not sufficient — it does not confirm stable gain dynamics. At $Q=20\text{M}/R=10\mu\text{m}$, 0.5mW passed the energy check but the CMT simulation showed operation $\sim 190\times$ above threshold, in a self-oscillating regime.

PART III — Open Problems Registry (Merged)

All outstanding items from all three source documents, listed explicitly rather than treated as settled.

1. Core Architecture Open Problems (from Physics Error Analysis, Appendix F of the original patent)

OP-1: Five-stratum SiPh yield characterisation

Three- and two-stratum SiPh platforms are demonstrated. Extending to five strata with ~203M PCM switches per layer at acceptable interlayer optical loss and wafer-scale yield has not been characterised. Critical questions: cumulative interlayer insertion loss across five strata, and yield of vertical optical vias at required density.

OP-2: Distributed degenerate GaP PSA pump phase-locking at scale

Degenerate PSA in individual waveguides is demonstrated. The open problem is maintaining phase coherence of the degenerate pump across millions of spatially separated GaP zones spanning a multi-tile datacenter fabric. Four passive mitigation strategies exist (GaP routing isolation, segmented island growth, global vertical pump illumination, 100µm scale limit), but formal experimental characterisation of residual phase error budget across a full multi-tile fabric remains future work. (Note: this item's relevance is reduced now that the amplification stage is Si₃N₄-based, but it remains open for any future GaP reintroduction.)

OP-3: Formal multi-fault CRT detectability bounds

CRT provides strong visibility under single-residue faults. A formal adversarial model for simultaneous multi-tile faults, correlated failures (shared thermal events across adjacent tiles), and faults confined within a single residue domain (not crossing the CRT inconsistency boundary) has not been established. Closed-form detectability bounds under worst-case conditions remain open.

OP-4: CMOS backend throughput verification at 100 GHz input rate

The parallel pipeline solution (21 engines at 4.76 GHz) is the correct engineering approach, but the CMOS adder-tree depth at TSMC N3 under the specific CRT reconstruction algorithm must be formally verified to confirm 4.76 GHz timing closure. CRT engine complexity scales with the number of residue tiles k and the modulus set bitwidths. A gate-level synthesis pass (e.g. Synopsys Design Compiler with N3 PDK) is required before finalising the 21-pipeline claim with precision.

Priority Ordering (original)

Priority	Open Problem	Blocking?	Next Step
1 (Highest)	OP-2: PSA pump phase-locking	Yes, for datacenter scale (if GaP revisited)	On-chip phase distribution architecture design
2	OP-4: CMOS 4.76 GHz closure	Yes, for 100GHz claim	Gate-level synthesis with N3 PDK
3	OP-1: 5-stratum yield	Yes, for fabrication	Partner with imec / GlobalFoundries SiPh PDK
4 (Lower)	OP-3: Multi-fault CRT bounds	No (safety margin, not blocking)	Formal fault model analysis

2. GaP Amplifier Open Problems

See Part II-B §B.11 for full detail. Summary list:

- Optical intensity / pump depletion check at the consolidated 8,192-site amplifier output vs. GaP's TPA edge and gain-saturation behavior.
- Non-degenerate FWM detuning (pump-signal wavelength offset) not yet derived for $Q \approx 290,000\text{--}300,000$.
- Hybrid TFLN/SiN-host + GaP-evanescent-layer path unverified — no device has jointly demonstrated high host Q and high GaP modal overlap.
- GaP-on-insulator bend-loss vs. radius dataset at $R=5\mu\text{m}$ not independently confirmed.
- Pump routing/delivery architecture to 8,192 sites not yet designed.
- EO/thermal locking strategy for the pure-GaP ring not re-derived for this specific geometry.
- Foundry availability at $Q \approx 290,000\text{--}300,000$ via accessible MPW — currently unresolved; this is why the design-of-record is Si_3N_4 , not GaP.

3. Si_3N_4 / Fabrication Open Items — Not Yet Resolved

These are genuinely unverified. Listed explicitly rather than treated as settled.

#	Item	Why It Matters
1	imec's true D2D (not D2W) bonding pitch/yield	Largest unverified number in the fabrication chain — confirm directly with imec first
2	GST thermal survival through post-bond Cu-Cu anneal	Different risk from switching-pulse crosstalk; not yet checked
3	LIGENTEC MPW pricing ($\sim \$350/\text{mm}^2$)	Never independently verified against a live quote
4	LioniX "30% discount"	No written source found
5	Real vendor for heater/GST attachment cleanroom step	"AMD" was an unresolved placeholder
6	SCL Mohali / Dholera 2027 node timeline	Genuinely uncertain from public information
7	C2S eligibility for a non-university-affiliated project	Verbally confirmed; worth having in writing
8	Multi-jurisdiction export control review	India, Switzerland, Netherlands, Belgium — never formally reviewed as one package
9	Matched-filter receiver noise improvement (+2–4dB est.)	Real technique; not yet built into a simulation to confirm

PART IV — References (Merged)

Deduplicated across all three source documents. Grouped by subject area.

1. Timing / Wave-Pipelining

- Lischke, S. et al. (2021). “Ultra-fast germanium photodiode with 3-dB bandwidth of 265 GHz.” *Nature Photonics*, 15, 925–931. doi:10.1038/s41566-021-00893-w
- Steckler, J. et al. (2024). “Ge electro-absorption modulator bandwidth characterization at 100 GHz.” *IEEE Transactions on Electron Devices*, 71, 3417–3423.
- Saleh, B. & Teich, M. *Fundamentals of Photonics*, 3rd ed., Wiley, 2019 — Ch. 22 (wave-pipelining), §3.1 (transform-limited pulse bandwidth).

2. GaP Nonlinear Photonics & Amplifier Design

- Wilson, D. J. et al. (2020). “Integrated gallium phosphide nonlinear photonics.” *Nature Photonics*, 14, 57–62. doi:10.1038/s41566-019-0537-9
- Logan, D. F. et al. GaP-on-oxide ring resonators: $Q \sim 10^4$ with $\sim 400\%/W$ second-harmonic-generation conversion efficiency (Princeton/UW).
- Kuznetsov, A. et al. (2024/2025). Integrated gallium phosphide travelling-wave optical parametric amplifier: 5.55cm spiral waveguide, up to 35dB gain. (Also presented as OFC 2024, Th1D.3.)
- Xiao et al. OFC 2024, paper Th1D.4. GeSbS microresonator phase-sensitive parametric amplification, 31.5dB gain at 8.5mW pump. Note: this paper describes a GeSbS device, not GaP — a citation-mismatch correction made during the GaP investigation (Part II-B §B.8.2).
- Xia, F. et al. (2022). $\text{Ge}_{25}\text{Sb}_{15}\text{S}_{60}$ chalcogenide nonlinear index (n_2) characterization via Z-scan; GeSbS microring resonators, $R \approx 100\mu\text{m}$, intrinsic Q up to $\sim 2.3 \times 10^6 - 5 \times 10^6$.
- Kashyap, R. *Fiber Bragg Gratings*, 2nd ed., Academic Press, 2009 — Ch. 8 (chirped fiber Bragg grating pulse compression).
- Hwang et al. (2025). TFLN photonic crystal Fabry-Pérot resonator via PLACE fabrication, loaded $Q = 1.4 \times 10^6$.
- Peng et al. (2025). *Optics Letters*. Single-crystal AlN-on-sapphire ring resonator, record intrinsic $Q = 2.8 \times 10^6$ at 1550nm.
- Ding et al. (Jan 2026). arXiv preprint. Polycrystalline/optimized AlN racetrack resonators, $Q = 5.4 \times 10^6$ (near-IR), dispersion-engineered for AlN Dirac solitons.

3. Power Budget, Detector Margin & Noise Floor

- Miller, D. A. B. (2017). “Attojoule Optoelectronics for Low-Energy Information Processing and Communications.” *J. Lightwave Technol.*, 35(3), 346–396.
- Winzer, P. & Essiambre, R. (2006). *J. Lightwave Technol.*, 24(12) — transimpedance amplifier noise floor analysis for photonic receivers.
- Lischke, S. et al. (2021). *Nature Photonics*, 15, 925–931 — used again as the source of real detector responsivity (0.3 A/W at 265GHz) for the -17.2dBm Si_3N_4 -stage noise floor recalculation.
- arXiv 2209.05589 — cited and checked directly for the rejected “45dB single-ring” Si_3N_4 claim (Part II-C §C.3); real demonstrated result is 30dB at 9mW, with gain saturation onset at 100pW.

4. Laser Source (CW + TFLN Solution)

- Wang, C. et al. (2018). “Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages.” *Nature*, 562, 101–104. doi:10.1038/s41586-018-0551-y
- He, M. et al. (2019). “High-performance hybrid silicon and lithium niobate Mach-Zehnder modulators.” *Nature Photonics*, 13, 359–364. doi:10.1038/s41566-019-0378-6
- Zhu, D. et al. (2021). “Integrated photonics on thin-film lithium niobate.” *Advanced Photonics*, 3(2). doi:10.1117/1.AP.3.2.028002
- Liang, D. & Bowers, J. (2010). “Recent progress in lasers on silicon.” *Nature Photonics*, 4, 511–517. doi:10.1038/nphoton.2010.167
- Ren et al. (2020). *J. Semiconductors*, 41(3). doi:10.1088/1674-4926/41/3/032901 — 60% WPE CW laser source; correctly applies only to CW operation, not the mode-locked laser it was originally (mis)applied to.

5. CMOS Backend / Parallel Pipelines

- Vogel, C. (2005). “The impact of combined channel mismatch effects in time-interleaved ADCs.” *IEEE Trans. Circuits Syst. I*, 52(12).
- Möller, M. et al. (2003). InP HBT 100 Gbit/s DMUX (1:2 at 100Gbps, extendable to 1:N). *Electronics Letters*, 39(1).
- Liu et al. (2021). SiGe BiCMOS 100G+ optical receiver front-end. *Microelectronics Journal*, 113.
- TSMC Technology Symposium 2022 — N3 process node timing characteristics (public roadmap).
- Vaidyanathan, P. P. *Multirate Systems and Filter Banks*, Prentice Hall, 1992.
- IEEE 802.3bs Standard (400GbE) — 1:4 DMUX standard, publicly available.

6. Core JANUS Architecture References (Confirmed Valid, Unchanged)

- Wuttig, M., Bhaskaran, H. & Taubner, T. (2017). “Phase-change materials for non-volatile photonic applications.” *Nature Photonics*, 11. doi:10.1038/nphoton.2017.126
- Fang, Z. et al. (2022). *Nature Nanotechnology*, 17 — GST zero-static-power PCM switch in SiPh. doi:10.1038/s41565-022-01153-w
- Xia, Y. (2000). “CRT error amplification / RNS fault visibility.” *IEEE Signal Processing Letters*, 7(11).
- Wilson, D. J. et al. (2020). *Nature Photonics* — GaP-on-Si CMOS-compatible nonlinear photonics (also listed in §2 above).
- Skacel, S. et al. (2021). *IEEE ECTC* — 3D SiPh heterogeneous integration.

7. Verification Note

A note carried forward from the GaP amplifier document applies to this entire merged record: several figures (particularly the GaP citation-mismatch correction in Part II-B §B.8.2) originated from a citation that was later found to be mismatched to the wrong material system. Every reference above has been checked at least once against a live web search during the original design process; however, full bibliographic detail (exact author lists, volume/page numbers) was not independently confirmed for every entry, particularly the 2025–2026 preprints. Before this document is used in an external-facing submission, each reference should be re-verified against the original source.

— End of Consolidated Design Record —

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